Where 4-Bit Quantization Breaks Small Tool-Calling Agents, and Which Repairs Fix It

CMPE 295A · Advisor review · November 13, 2026

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M2 status · references, parity and cluster capacity · decisions for you · plan to M3 (Dec 4)

M2 is largely in place; four items remain

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models in BF16 + three quantized formats, KL-checked

3 / 6

BF16 references done (Qwen3 0.6B, 1.7B, 4B)

7,506

episode verdicts, all equal to the official evaluators

Done

- Clean-clone rerun under a second member's account (PASS)
- S1, S7 and S10 on the cluster; Slurm array runner with requeue
- BF16, Q4_0, Q4_K_M and Q8_0 for six models, with KL vs BF16
- CUDA-vs-CPU parity test (FR-E17) and throughput pilot (NFR-5)
- τ^2 simulated user as its own cluster job
- Inclusion filter, codebook v0 (+2 classes), mixed-effects R script

In progress

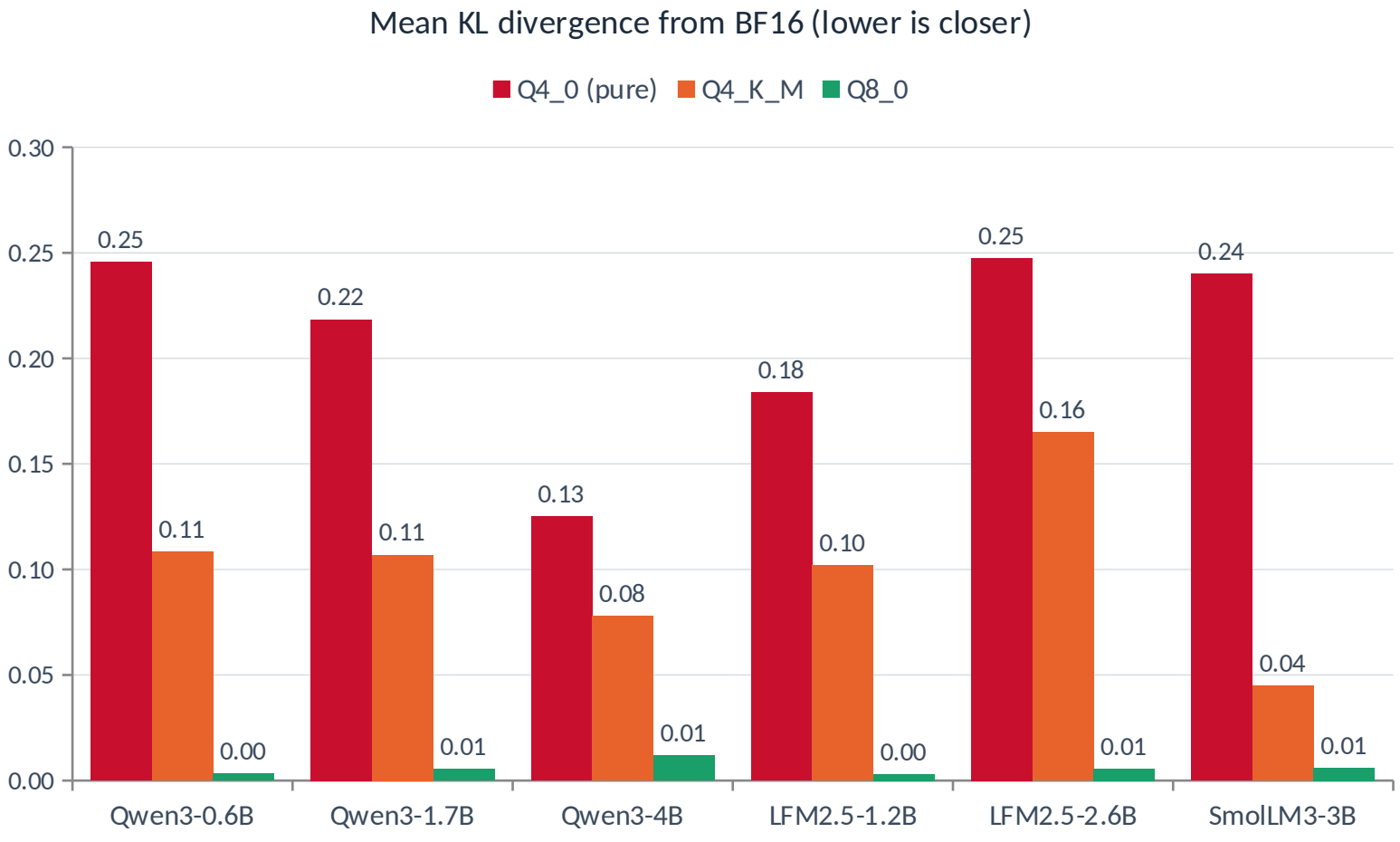
- SmolLM3 and LFM2.5 references: prompt handlers first
- OSF pre-registration: draft waits on today's decisions

Open

- Human labels for the 30-step rater sample
- Merge m2 and m2-phase1 into main (PR in review)

M2 exit as planned on Oct 30: BF16 references for every model, two members' own cluster runs, codebook v0, branches merged. Sources: docs/cluster.md, docs/m2_decisions.md, results/m2/ (branch m2-phase1).

Six models, four formats: Q8_0 stays close to BF16, 4-bit does not



Same top token as BF16

Q8_0: 94.8–96.2%

Q4_K_M: 81.1–88.9%

Q4_0: 74.7–85.7%

Cite KLD, not perplexity

LFM2.5-2.6B Q4_K_M lowers WikiText perplexity (ratio 0.86, rechecked with 200 chunks) while its KLD is 0.165.

llama-perplexity --kl-divergence vs each BF16 file; WikiText-2 raw test, ctx 512, 50 chunks, CUDA on a P100. Q4_K_M ≈ 5.0 bits per weight, Q4_0 ≈ 4.5. results/m2/kl_*.json, models_manifest.json.

BF16 references: even 16-bit agents rarely finish a task

Model	BFCL base	miss_func	miss_param	τ^2 airline	τ^2 retail	τ^2 telecom
Qwen3-0.6B	4.5% (6.0)	2.0%	1.2%	0%	1.9%	0%
Qwen3-1.7B	6.2%	2.2%	8.5%	0%	3.6% (1.9)	0%
Qwen3-4B	12.2%	9.5%	11.0%	3.8% (0)	8.2% (10.6)	1.1%

Episode success over 4 sampled trials; greedy in brackets where it differs by more than 1 point. 501 tasks per model: BFCL multi-turn 3 × 100 + τ^2 write tasks (airline 26, retail 104, telecom 71). 2,505 episodes each.

So the primary outcome is per step (decision 2)
With success near zero, an episode-level comparison of 4-bit vs BF16 has almost nothing to compare. Failures per step exist in every episode, solved or not; episode success stays as the secondary outcome.

7,508 / 7,515

episodes finished; 7 excluded by the empty-reply rule

4 + 1

sampled trials (T 0.7, top-p 0.8, top-k 20) + 1 greedy pass

40,960

token context, 4,096 new tokens per step (bfcl parity)

Runs ref_qwen3_0p6b/cd314594fb, ref_qwen3_1p7b/c456da2aef, ref_qwen3_4b/1c7b26d4e7 (2,505 specs each). Development data. docs/cluster.md, M2 results 2026-11-09.

Our scoring matches the official evaluators on every episode

7,506 / 7,506

verdicts identical to bfcl-eval and τ^2-bench when every finished reference episode is re-scored from its logged trajectory (FR-E2: PASS)

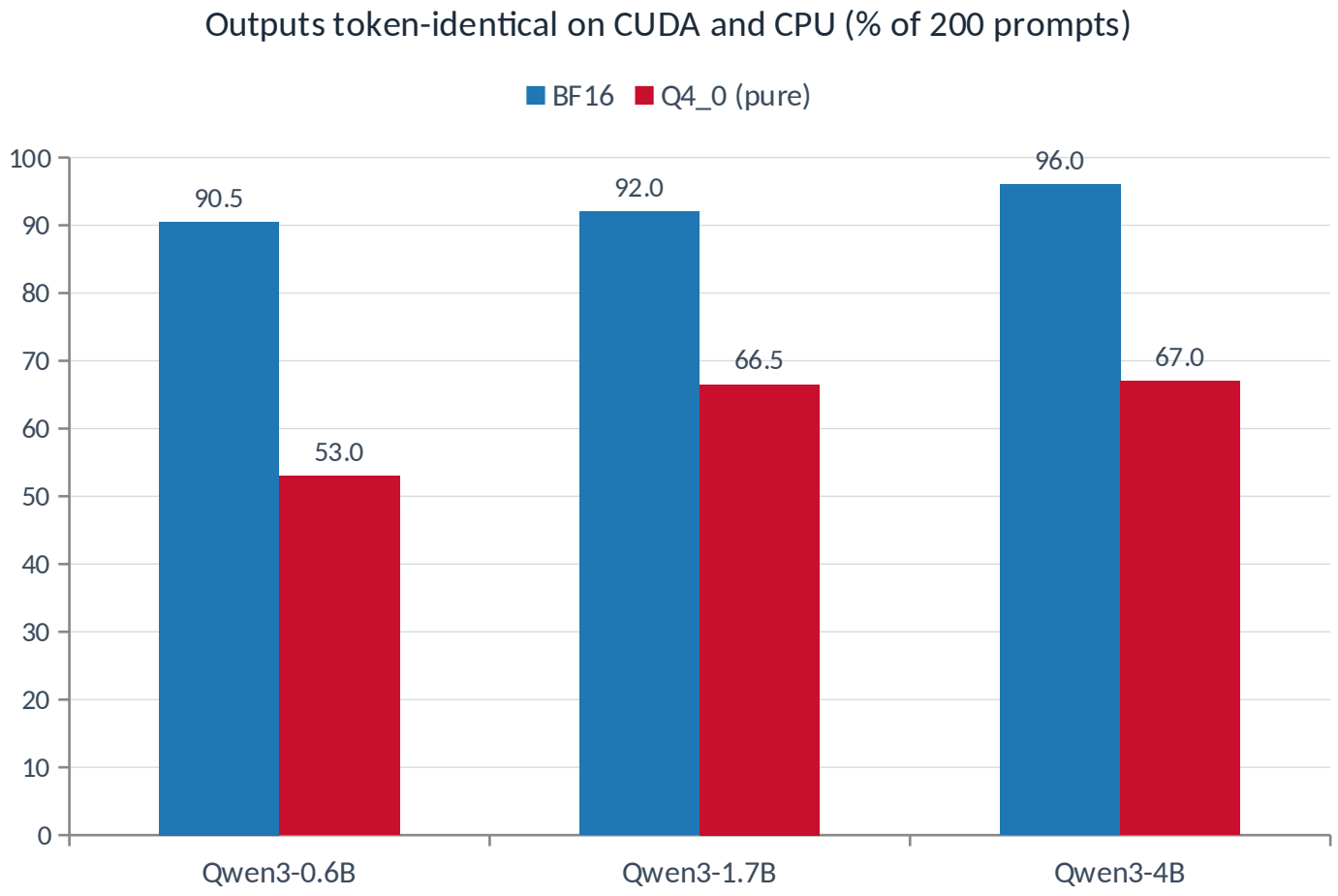
BFCL: 4,500 of 4,500
Re-scored through bfcl-eval's own path for a local Qwen3 FC model: its response parsing, decode_execute and multi-turn checker.

τ^2-bench: 3,006 of 3,006
Re-scored with τ^2's evaluate_simulation on the saved simulations.

Two episodes not re-scorable
Two τ^2 episodes ended in a context overflow (one agent, one simulated user) and have no simulation to re-score; they are scored as failures without the evaluator.

scripts/evaluator_agreement.py, CPU job 110190; results/m2/evaluator_agreement.json. bfcl-eval's API-client imports are replaced with inert stand-ins (listed in the JSON); multi-turn scoring never calls them.

CUDA and CPU agree less at 4 bits, so the backend stays a factor



100%

of prompts tokenize identically on both backends

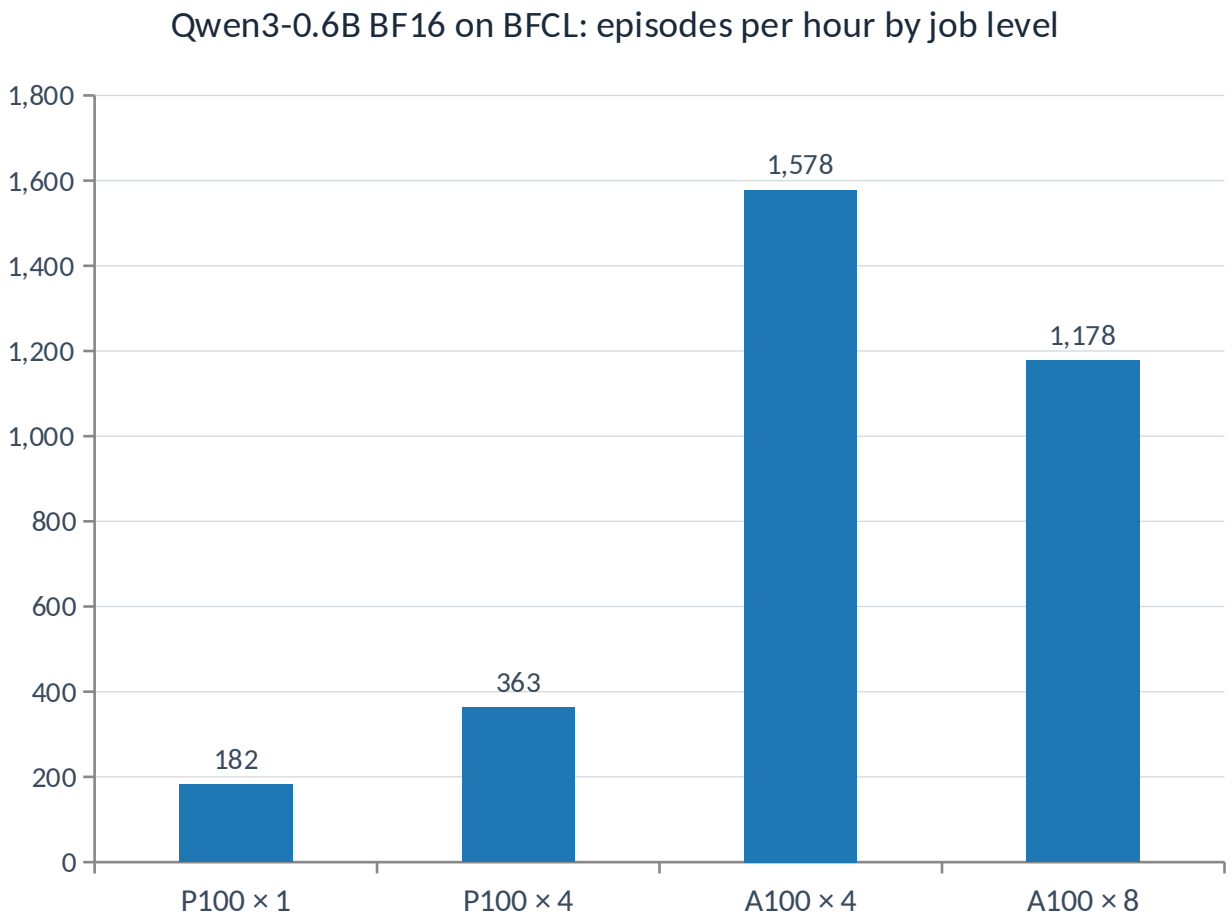
97–100%

same call-or-reply decision (Q4_0 97.0–98.5%)

Pooling rule not met (decision 7)
We pool backends only at ≥ 99% identical outputs. No model reaches it, so backend is a recorded factor in every RQ1 analysis.

FR-E17: 200 fixed prompts per model (Oct 28 BF16 episodes replayed), greedy, ctx 40,960, cache off, same GGUF file. CUDA on P100 (4B BF16 on H100), CPU on cpuqs, 32 CPUs. results/m2/parity_qwen3-*.json.

The 2-node limit per user, not the GPUs, sets our pace



Level	Jobs ran at once	Median episode	Queue wait
P100 × 1	1	15.5 s	54 min
P100 × 4	2	16.5 s	65 min
A100 × 4	2	2.4 s	167 min
A100 × 8	2	2.3 s	175 min

NFR-5 (≥ 8 concurrent jobs) is not met
QoS gpu3nodes allows 2 nodes per user and the A100 nodes are shared; episodes per hour (≥ 120) passes easily. Plan: 3–7 concurrent A100 jobs. Ask: can you request a higher node limit, or may shares run from our other accounts?

scripts/throughput_submit.sh, 24 episodes per job; arrays 109478–109481. results/m2/throughput.json. A100 nodes cs001 and cs002 excluded (bad GPUs).

What the main grid can detect at 2,000 episodes per configuration

Failure class	Rate per step (0.6B / 1.7B / 4B)	Detectable?
any failure	30% / 27% / 23%	yes, all models
exec_error	7.2% / 10.4% / 6.6%	yes
tool_error_payload (new)	9.4% / 8.0% / 6.2%	yes
unknown_tool	7.0% / 5.5% / 8.5%	only at low clustering
missing_call	1.0% / 1.4% / 0.9%	borderline
missing_call_direct_answer (new)	2.9% / 0.5% / 0.6%	0.6B only
invalid_args	2.7% / 0.9% / 0.2%	no
malformed, forced_quit, truncated	< 0.3% each	no: pool or explore

Tasks the BF16 reference solves
0.6B: 17 · 1.7B: 33 · 4B: 73 of 501
All under the registered 120, so success-based inclusion applies only to the secondary analysis (decision 11).

Proposal for the pre-registration
Confirmatory: any failure, exec_error, tool_error_payload. Exploratory or pooled: the rarer classes.

Registered test: 4-bit share of events 0.5 vs 0.6, α = 0.05/10, 80% power → 330 events at design effect 1. Verdict uses design effects between task and episode clustering. results/m2/power_table.json, inclusion_qwen3-*.json.

Eleven decisions we need before we freeze the analysis plan

Change the pre-registration

2 Step-level failures primary; episode success secondary

11 Success-based inclusion only for the secondary analysis; all 501 tasks for step-level tests

3a Add "missing call with a direct answer" to codebook v0

3b Own class for a tool that returns an error payload

8 Depth bins by step index: 0–1, 2–3, 4–6, 7+

Fix the setup

1 Context 40,960, 4,096 new tokens per step (bfcl parity); update FR-E3

4 τ^2 simulated user as a cluster job; drop the GCP L4 request

5 References: 4 sampled trials + 1 greedy pass

6 τ^2: only tasks whose gold actions write (201 of 278)

7 Backend a recorded factor (parity < 99%)

Schedule and scope

9 Keep M4 on Dec 18

10 Vision-language extension: optional 295B scope, go/no-go at M6

If you already decided any of these on Oct 30, that decision stands; tell us and we will update the record before we file the pre-registration.

Proposals by Ekant in docs/m2_decisions.md (2026-11-08), each awaiting your confirmation.

Plan to M3 (Dec 4): damage map v1

Ekant

- 4-bit damage runs as Slurm arrays (to Dec 4)
- R1 grammars per tool-call format
- Second labeler for the 300 pilot steps

Vritika

- 300 labeled pilot steps and κ → codebook v1
- Damage map v1 (Dec 4)
- OSF filing once you confirm the decisions

Yashashav

- R2 variants by Nov 22: Q5_0, Q6_K; layer-selective 5.5 bits
- IQ4_XS with an importance matrix; KL on CPU and CUDA
- QAD data (critical path); adjudicate label disagreements

M3 exit: Q4_0, Q4_K_M and IQ4_XS of every model, KL-checked on both backends; grid run as Slurm arrays; codebook v1 with κ; step-type and depth annotations.

Open for the team: the SmolLM3 and LFM2.5 references need prompt handlers first; we will settle who adds them this week.

Workbook 2 sprint plan (Sprint 6, Nov 9–22; Sprint 7, Nov 23–Dec 6), work packages H1, H2, I, J1, K.

Dec 4: four Canvas items, and the demo we propose

Report: first two chapters

Group · locks at the deadline. Ch. 1 Project Overview, Ch. 2 Architecture. Draft to you Tue Nov 24.

Early prototype demo

Group · graded by you. Proposal: a 5–10 min recording of one run spec → Slurm array → step log → classifier → damage-map figure.

Project contribution

Individual · each of us, from the git history and meeting notes.

Meeting log 2

Group · link to the /meetings folder (Oct 1–29 notes).

Nov 24
Chapter draft to you

Dec 3
Record the demo

Dec 4
Submit all four

Dec 11
Project Expo

Canvas assignments 7852172, 7852164, 7852170, 7852168; all due Fri Dec 4, 11:59 PM. Project Expo Fri Dec 11, 10:30 AM.

What we would like from you today

1. Confirm the decisions, above all 2 and 11 (they change the pre-registration)
2. OSF filing: under whose account, and may we list you as a contributor?
3. Cluster: can you request a higher node limit, or may grid shares run from our other accounts?
4. Demo format for Dec 4: recording or live?
5. Next meeting: Nov 27 is the day after Thanksgiving. Mon Nov 30 or Wed Dec 2, 1:00 PM?
6. Canvas still lists our title as "KV Cache Behavior…": how should we get it changed?

Appendix: cluster checks behind M2

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S7: training steps with losses identical to the uninterrupted run after a walltime cut and requeue, across two nodes

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S10: episodes after every array task was cut and resumed; no missing or duplicate episodes

PASS

Clean clone under Ekant's account: experiment hash 3e3a25e70f, 253 specs equal

- S7 under a 10-minute limit: B cut at step ~368, resumed from the step-365 checkpoint; max loss difference 0.0. Peak GPU memory 18.8 / 19.9 GB of 40 GB.
- S10 needed one resubmission: R1 with the forced grammar on Q4_0 used all 5 requeues with 3 episodes left; the second array ran only those 3. Size slurm.time from the slowest configuration.
- Bad nodes: A100 cs001 (GPUs 1–3 fail CUDA init) and cs002 are excluded from A100 jobs. Requeue probe: JobRequeue = 1, USR1 60 s before the limit.

results/m2/s7_compare.json, s10_check.json, s10_check_first_submission.json, clean_clone_ekant.json; docs/cluster.md.